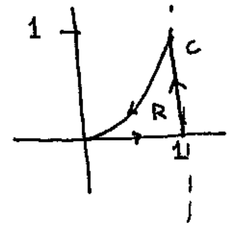
Prob 1

(a) $\vec{F} = 2xy^3\,\vec{i} + 4x^2y^2\,\vec{j}$



$$\text{curl}\,\vec{F}\cdot\vec{k} = 8xy^2 - 6xy^2 = 2xy^2$$

$$\text{div}\,\vec{F} = 2y^3 + 8x^2y$$

(i) $$\oint_C \vec{F}\cdot\vec{T}\,ds = \iint_R 2xy^2\,dA = \int_0^1\int_0^{x^3} 2xy^2\,dy\,dx$$

$$= \int_0^1 \frac{2}{3}xy^3\Big|_0^{x^3}\,dx$$

$$= \int_0^1 \frac{2}{3}x^{10}\,dx$$

$$= \frac{2}{33}x^{11}\Big|_0^1 = \frac{2}{33}$$

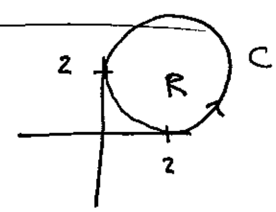
(ii) $$\oint_C \vec{F}\cdot\vec{N}\,ds = \iint_R 2y^3 + 8x^2y\,dA = \int_0^1\int_0^{x^3} 2y^3 + 8x^2y\,dy\,dx$$

$$= \int_0^1 \frac{1}{2}y^4 + 4x^2y^2\Big|_0^{x^3}\,dx$$

$$= \int_0^1 \frac{1}{2}x^{12} + 4x^8\,dx$$

$$= \frac{1}{26}x^{13} + \frac{4}{9}x^9\Big|_0^1 = \frac{1}{26} + \frac{4}{9}$$

(b) $\vec{F} = [4x-2y,\ 2x-4y]$



$$\text{curl}\,\vec{F}\cdot\vec{k} = 2 + 2 = 4$$

$$\text{div}\,\vec{F} = 4 - 4 = 0$$

(i) $$\oint_C \vec{F}\cdot\vec{T}\,ds = \iint_R 4\,dA = 4\cdot\text{Area}(R) = 4(\pi 2^2) = 16\pi$$

(ii) $$\oint_C \vec{F}\cdot\vec{N}\,ds = \iint_R 0\,dA = 0$$

Prob 2

$$\frac{1}{2}\oint_C -y\,dx + x\,dy \overset{\text{Green's Thm}}{=} \frac{1}{2}\iint_R 1-(-1)\,dA = \frac{1}{2}\iint_R 2\,dA$$

$$= \iint_R 1\,dA$$

$$= \text{Area}(R)$$

Prob 3

$$\underbrace{\oint_C ky\,dx + hx\,dy}_{\circledast} \overset{\text{Green's Thm}}{=} \iint_R (h-k)\,dA = (h-k)\iint_R dA$$

$$= (h-k)\,\text{Area}(R)$$

The value of $\circledast$ is the area of R scaled by the constant h-k